

1. Lifecycle Methodology

The design of the e-health.albania application followed a strict User-Centered Design (UCD) lifecycle methodology. This ensured that complex clinical datasets, state tracking, and strict security compliance were unified into an optimal, high-velocity workflow tailored to medical professionals.

2. Process Execution Stages

Stage 1: Requirement Extraction & Role Mapping

Extracted exact workflows from all 20 clinical scenarios and 50 core functional requirements. Tasks were carefully divided into operational boundaries specific to defined user accounts (Doctors, Nurses, Receptionists, Cashiers, Lab Technicians, and System Administrators).

Stage 2: Operational Task Analysis & Click Optimization

Analyzed user movement patterns within clinical environments to optimize mouse paths. The visual layout was engineered to guarantee compliance with **NFR-14**, establishing that any core role-specific clinical or administrative action can be successfully executed within a maximum of three clicks from that user's home dashboard.

Stage 3: High-Density Layout & Grid Wireframing

Created structural desktop-optimized wireframes focusing on tabular grids and panel split-views. Instead of deploying web-style whitespace, the screens prioritize data optimization, maximizing parallel tracking of client files, clinical metrics, and diagnostic orders without requiring users to continuously jump across separate workspace sheets.

Stage 4: JavaFX Architecture Decomposition & Separation of Concerns

Decomposed final visual screens into strict, reusable JavaFX controls using the Model-View-Controller (MVC) pattern. Form fields utilize custom reactive properties to validate text input, date ranges, and national identifiers instantly at the view layout boundary prior to committing state changes to the business layer.

3. Interface Evaluation Criteria

The application views are systematically validated against strict performance and accessibility metrics prior to packaging:

- **Usability Benchmark (NFR-13):** Form layouts must be structured logically so that a newly trained receptionist can complete a new patient registration sequence inside an intake window of five minutes.
- **Search Responsiveness (NFR-02):** Asynchronous JavaFX Task bindings are implemented on all search layouts to guarantee patient queries return clear visual results in less than two seconds without freezing the primary GUI engine.
- **Accessibility Conformance (NFR-15):** All layouts provide direct keyboard mnemonics, intuitive tab navigation order mappings, and meet high-contrast WCAG 2.1 Level AA digital standards.